

Creating a Prototype for a Data Anonymization Service

MASTER THESIS

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Declaration of Originality

I confirm that the submitted thesis is original work and was written by me without further assistance. Appropriate credit has been given where reference has been made to the work of others. The thesis was not examined before, nor has it been published. The submitted electronic version of the thesis matches the printed version.

Erlangen, 30 July 2026

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Abstract

General instructions: If the thesis is written in German, this section contains the English translation. The abstract is 100–200 words maximum.

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Acronyms

DSGVO Datenschutz-Grundverordnung

GDPR General Data Protection Regulation

FLR functional legal requirements

FSR functional system requirements

FIR functional integration requirements

NFLR nonfunctional legal requirements

NFSR nonfunctional system requirements

NFIR nonfunctional integration requirements

LLM large language model

UUID universally unique identifier

BDSG Bundesdatenschutzgesetz

1 Introduction

Deriving actionable insights from vast datasets is fundamentally driving innovation across diverse industries, from healthcare analytics to financial modeling. Yet, this heavy reliance on data accumulation presents a critical dichotomy. On one hand, organizations are legally and ethically mandated to protect individual privacy under frameworks such as the General Data Protection Regulation (GDPR) and emerging global standards. On the other, they remain wholly dependent on utilizing that same data to train machine learning models and perform complex statistical analyses.

In the specialized field of software engineering, this conflict is centered on the vast amount of metadata generated by development teams through their daily interactions with tools like GitHub and Jira. Effectively harnessing this information to drive productivity while respecting privacy requires innovative frameworks that can navigate the complexities of high-dimensional data.

The MECOIS project, a research initiative from the University of Erlangen, addresses the "Big Data" challenge within the specific domain of software engineering by transforming the vast metadata generated during development into actionable intelligence. Software development teams leave a continuous trail of activity-based artifacts in repositories such as GitHub and Jira, creating a high-dimensional dataset that is traditionally difficult to interpret due to overlapping metrics, complexity, and incompatible value ranges. MECOIS solves these issues by applying statistical methods—specifically the creation of a composite index—to group correlating metrics into key dimensions. This data-driven approach provides an objective Productivity Index that allows organizations to compare performance across teams and projects, identifying early-stage bottlenecks like miscommunication or resource gaps while uncovering the best practices of high-performing teams. (Riehle, 2025)

To address the tension between data utilization and privacy, this thesis proposes a prototype for a dedicated anonymization service designed to process the metadata collected within the MECOIS framework. While MECOIS extracts high-dimensional, activity-based artifacts from platforms such as GitHub and Jira to

1. Introduction

provide deep insights into team productivity, the granular nature of this data — including timestamps, commit details, and developer identities — presents significant privacy risks. The proposed service aims to anonymize this data in strict accordance with the GDPR, ensuring that the trail of artifacts used to calculate the MECOIS Productivity Index is stripped of personally identifiable information. By integrating this anonymization layer, the project seeks to enable the MECOIS's mission of empowering software developers through data-driven analysis while maintaining a privacy-first approach that meets modern legal and ethical standards.

2 Key Terms and Definitions

2.1 General Data Protection Regulation

The Datenschutz-Grundverordnung (DSGVO) (often also abbreviated to DSGVO) is the German name for the GDPR, which is a European Union regulation designed to ensure the protection of personal data and the free flow of data in Europe. It has been mandatory in all EU member states since May 25, 2018, and applies equally to private companies, public authorities, associations, and individuals. The GDPR strengthens the rights of EU citizens, as they now enjoy the same legal certainty in all member states and can decide how their own data is processed. For companies, it simplifies cross-border activities through uniform rules of conduct and the so-called “one-stop-shop” procedure, but also provides for substantial fines in the event of violations. While it is directly applicable in all EU member states, individual countries still have their own national data protection laws to supplement it. (für den Datenschutz und die Informationsfreiheit [BfDI], 2025)

Some parts of the GDPR can be changed or complemented by the states. Germany added the Bundesdatenschutzgesetz (BDSG) (English: Federal Data Protection Act) to the DSGVO to expand the mandatory regulations.

2.2 Pseudonymization

According to Article 4 No. 5 of the GDPR, pseudonymization is defined as the processing of personal data in such a manner that it can no longer be attributed to a specific data subject without the use of additional information. This additional information, often referred to as a “key” for re-identification, must be kept separately and be subject to rigorous technical and organizational measures to ensure that the personal data is not assigned to an identified or identifiable natural person. Crucially, pseudonymization is characterized as a processing operation rather than a static state, involving the active transformation of clear-text data into pseudonyms to mitigate risk while maintaining the utility of the

dataset. Because the underlying connection to a natural person is merely obscured rather than permanently severed, pseudonymized data remains classified as "personal data" and thus stays fully within the legal scope and obligations of the GDPR. (Members of the "Deutsche Gesellschaft für Medizinische Informatik et al. [GMDS], 2024, pp. 8, 21, 71) (Schwartzmann et al., 2022, p. 14)

2.3 Anonymization

In contrast, anonymization is the process of transforming personal data into anonymous information so that the data subject is no longer identifiable. While not explicitly defined within the text of the GDPR itself, it is recognized as a process that results in data that does not relate to an identified or identifiable natural person. A successful anonymization is fundamentally irreversible; once the process is complete, a re-identification of the individual must be impossible for the controller or any third party. Within the academic and legal discourse, a distinction is often made between "absolute anonymization," where re-identification is factually impossible for everyone, and "factual" or "relative anonymization," where re-identification would require a disproportionate effort in terms of time, cost, and labor. Once data is effectively anonymized, it is no longer considered personal data, and the requirements of the GDPR cease to apply. (GMDS, b., BvD, 2024, pp. 1, 9–10, 13, 21, 68, 73) (Schwartzmann et al., 2022, pp. 3, 16–19, 23)

2.4 Difference Pseudonymization and Anonymization

The fundamental difference between the two terms lies in the reversibility of the process and the subsequent legal classification of the data. Pseudonymization is a reversible procedure where the link between the data and the individual is preserved via a separate key, meaning the data retains its character as "personal data". Anonymization, however, aims for an irreversible state that permanently breaks this link, thereby removing the data from the regulatory framework of the GDPR. Furthermore, pseudonymization requires the controller to implement strict silos and access controls to manage re-identification information, whereas anonymization requires the permanent deletion or sufficient generalization of all direct and indirect identifiers to ensure that no such "key" exists to restore the original identity. (GMDS, b., BvD, 2024, pp. 8, 13, 31, 68) (Schwartzmann et al., 2022, pp. 3, 19)

3 Literature Review

This section reviews existing scientific publications to establish a baseline for data anonymization services within software engineering productivity metrics.

3.1 Data Protection Laws

Finding publications about this topic is very difficult. There are a lot of sources about the DSGVO and anonymization and their applications. But most of them are too broad for this thesis or focus on surveillance tools.

Christl (2021) provides a systematic mapping of digital surveillance and algorithmic management across diverse industries. Through case studies like Amazon and Zalando, he documents a trend toward "digital Taylorism," characterized by real-time tracking that intensifies work and reduces employee autonomy. Ultimately, the study shows how these technologies reinforce power imbalances between employers and labor through pervasive data collection and automated control.

The standard reference work on employee data protection is Däubler (2017). Written in a practical style and accessible even to non-specialists, this handbook explains the entire field of data protection law. It focuses on employee data protection and the participation rights of works councils and staff councils. It also addresses the challenges posed by AI that are already apparent today, such as those arising in the hiring process or when working with ChatGPT. Another key focus is on the diverse means available to ensure that data protection is effectively enforced in practice. These range from data backup and the activities of the company data protection officer and the supervisory authority to the imposition of substantial fines and numerous claims for damages. The now very extensive case law of the European Court of Justice (ECJ) — which has set many new precedents — is analyzed in relation to these and numerous other questions. As a rule, this case law strengthens the protection of individuals, which is also of great significance for the scope of action available to works councils and staff councils. This shift has not yet been fully recognized everywhere.

In the context of data protection and research, the operationalization of de-

identification procedures under the DSGVO is addressed by two central German guidelines that provide both technical and legal frameworks. Schwartmann et al. (2022) establishes a structured procedural model that distinguishes between theoretical "absolute anonymity" and the more practically relevant "factual anonymization," where re-identification is considered impossible due to the disproportionate effort required in terms of time, cost, and manpower. Complementing this, GMDs, b., BvD (2024) focuses specifically on the sensitive domain of health data under Art. 9 GDPR, highlighting the necessity of aligning national definitions with the broader European legal framework provided by Directive (EU) 2019/1024. While Schwartmann et al. (2022) categorizes technical methods—such as randomization (e.g., differential privacy) and generalization (e.g., k-anonymity)—into specific application classes like AI algorithm training or software testing, GMDs, b., BvD (2024) provides detailed risk assessments for specific attack scenarios, including "singling out," inference, and linkage attacks. Both sources emphasize that anonymity is not a static state but a dynamic process that must be continuously evaluated against the evolving "state of science and technology". Furthermore, they converge on the legal requirement that both anonymization and pseudonymization constitute "processing" activities under the GDPR, therefore requiring a valid legal basis and remaining subject to the controller's accountability obligations.

3.2 Server Design

TODO

3.3 Identity Management

The MECOIS project uses the tool SortinhHat from the toolset GrimoireLab by CHAOSS for managing identities.

Dueñas et al. (2021) present GrimoireLab, an integrated and modular open-source toolset developed to facilitate the comprehensive retrieval, storage, and visualization of data from software repositories. This framework addresses a critical gap in software analytics by providing automated support for approximately 30 different data sources, including version control systems like Git, issue trackers such as Jira and Bugzilla, and communication platforms like Slack and mailing lists. Central to its architecture is a multi-stage pipeline that employs specialized components: Perceval for uniform data retrieval via JSON documents, Graal for detailed source code analysis, and SortingHat for advanced identity management and cross-platform contributor merging. Furthermore, GrimoireLab utilizes a distinct storage model that maintains "raw" copies of original data alongside "enriched" indexes optimized for visualization, a strategy that signif-

icantly enhances data maintainability, traceability, and research reproducibility. By providing a mature, field-tested framework for both academic research and industrial application, the authors demonstrate that GrimoireLab reduces the manual effort required for empirical software engineering studies while enabling the longitudinal tracking of developer activity and project health.

SortingHat is the specialized identity management component within the GrimoireLab framework, specifically designed to address the challenge of contributor fragmentation across diverse software repositories. Because developers frequently utilize different usernames, aliases, and email addresses across platforms such as Git, Slack, and Jira, SortingHat provides the essential infrastructure to unify these disparate identities into unique, individual profiles. The component operates by maintaining a dedicated relational database that stores identity data, affiliation tags, and the original source of each identity to ensure research traceability. To maintain high data quality, SortingHat utilizes a conservative algorithmic approach to merging—avoiding false positives caused by shared generic accounts—which is further complemented by HatStall, a web-based interface that allows researchers to manually curate identities and manage organizational affiliations. This process of identity unification is fundamental to the broader GrimoireLab pipeline; it allows the analytics engine to enrich data points with a unique contributor universally unique identifier (UUID), thereby enabling the longitudinal tracking of developer trajectories and cross-platform activity metrics that would otherwise remain siloed. (Dueñas et al., 2021)

3. Literature Review

4 Requirements

The purpose of this chapter is to define the requirements necessary to facilitate a GDPR-compliant productivity benchmarking system. ‘ISO/IEC/IEEE International Standard - Systems and software engineering–Vocabulary’ (2017) splits requirements into functional and nonfunctional ones. Functional requirements define what a product must do and nonfunctional ones declare how the system should operate. Furthermore, the requirements can be classified into legality, system and integration. The service must abide to relevant data protection laws. System requirements specify how the service itself will be designed according to the author. The MECOIS project itself sets some integration demands in its contributing guidelines (‘MECOIS Github Repository’, 2026), that must be satisfied.

4.1 Functional Requirements

The table 4.1 specifies what the anonymization must do. They are sorted by functional legal requirements (FLR), functional system requirements (FSR) and functional integration requirements (FIR).

The Data Anonymization requirement FLR-1 is defined very vague by design. There can be many ways to anonymize data according to the DSGVO. Chapter 6 will compare and present different approaches. It stands in direct conflict with FSR-1 Preserve Data. The aim is to find a compromise between these two.

4.2 Nonfunctional Requirements

The table 4.2 specifies how the anonymization service must operate. They are sorted by nonfunctional system requirements (NFSR) and nonfunctional integration requirements (NFIR). There are no nonfunctional legal requirements (NFLR), because it is evaluated in chapter 6, how the legal requirements will be satisfied.

Table 4.1: Functional requirements

ID	Requirement	Description
FLR-1	Data Anonymization	The service must follow the GDPR which is defined by the DSGVO in Germany.
FSR-1	Preserve Data	The service must preserve all available Data.
FSR-2	independent Deployment	The service can be deployed and run independent from the rest of the MECOIS project.
FIR-1	Seamless Integration	The service must integrate into the project with as few changes as possible to existing features.

Table 4.2: Nonfunctional requirements

ID	Requirement	Description
NFSR-1	Containerized Service	The service must run in a docker container.
NFIR-1	Programming Language	The Backend of the MECOIS project is written in python3. Therefore, this project must also use python3. ('MECOIS Github Repository', 2026)
NFIR-2	Licence Policy	The contributing guidelines ('MECOIS Github Repository', 2026) state: "You are allowed to use any package that can help you do your work, as long as it is not a copyleft or proprietary license. However, it is still a good idea to ask beforehand, so you do not put in unnecessary work."
NFIR-3	Styleguide and Clean Code	The contributing guidelines ('MECOIS Github Repository', 2026) state: "We follow the PEP8 standard (with minor exceptions like allowing longer line limits). You can run the custom linter using: pylint. Additionally, your code must follow these rules before being merged into the main branch: • Remove comments that are not necessary for understanding your code • Remove unnecessary print statements • Make sure to update and provide the DocString description of the methods you write. Make sure to have it adjusted to your use-case if you copy method-stubs from other part of the code"

4. Requirements

5 Architecture

The MECOIS setup consists of 3 separate systems, that communicate via HTTP (see Figure 5.1). The systems are separated to protect sensitive information.

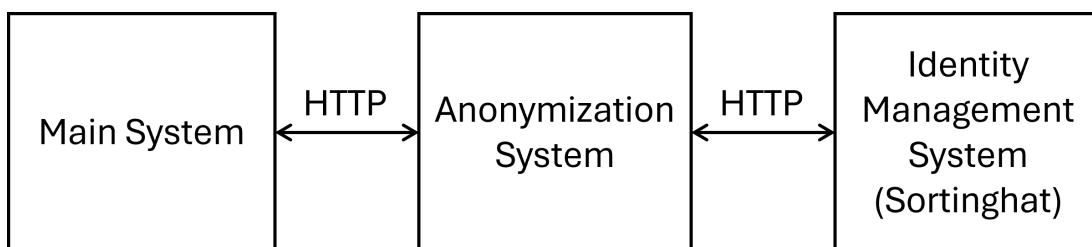


Figure 5.1: Basic System Architecture

5.1 Main System

The main systems is responsible everything but the identity management and therefore has many components.

5.1.1 Components

Figure 5.2 displays an overview of all components. (‘MECOIS Github Repository’, 2026)

Sources

The sources define the origin for the data. Currently we are using Git (Github and Gitlab), Confluence and Jira as source. For Example from Github we pull among other commits, issues and pipelines.

Infrastructure

The infrastructure provides the database for the project. The data is stored in a datalake, which is a repository, where data is stored as-is without further

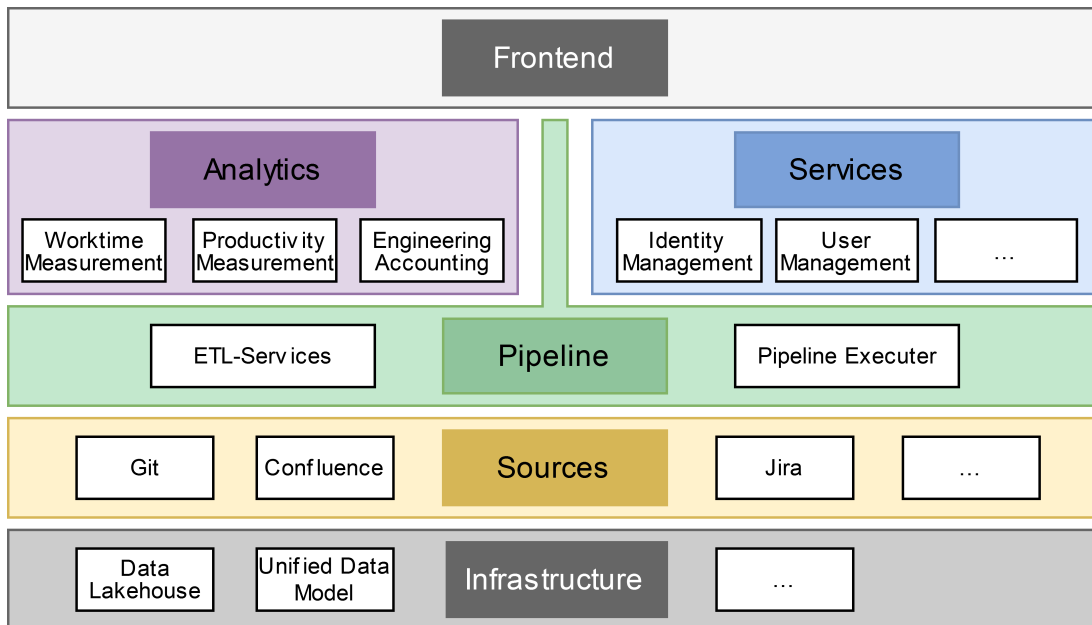


Figure 5.2: Main System Components (‘MECOIS Github Repository’, 2026)

structure (‘What is a data lake?’, 2026). Each set of data passes through 4 Stages: raw, bronze, silver, gold.

Pipeline

The pipelines transform the data between the stages.

Analytics

In the last step the data is used to provide multiple analytics for customers. Examples for such analytics are:

- Calculating the Costs of Inner Source Collaboration by Computing the Time Worked (Buchner & Riehle, 2021)
- Estimating Software Contribution Time from Git and GitHub Metadata (Stenzel, 2026)

Services

Services are helper modules for pipelines and analytics. Such helper modules are:

- AI: a framework for communicating with a large language model (LLM).
- config handling: Parses configuration files for pipelines.

Frontend

The frontend provides a Dashboard for the user-facing components of MECOIS.

Orchestrators

The orchestrators can be used to run the pipeline. The set up the configuration and run the pipeline for the selected sources.

5.1.2 Dataflow

After pulling the source the raw data is stored exactly as provided as JSON. In the datalake the Data is processed multiple times and the states are called bronze, silver and gold. The processing is done by the pipelines.

Each pipeline has 3 steps:

1. Read the data from the datalake
2. transform data
3. write the transformed data to the datalake

Figure 5.3 displays an overview of all components. (‘MECOIS Github Repository’, 2026)

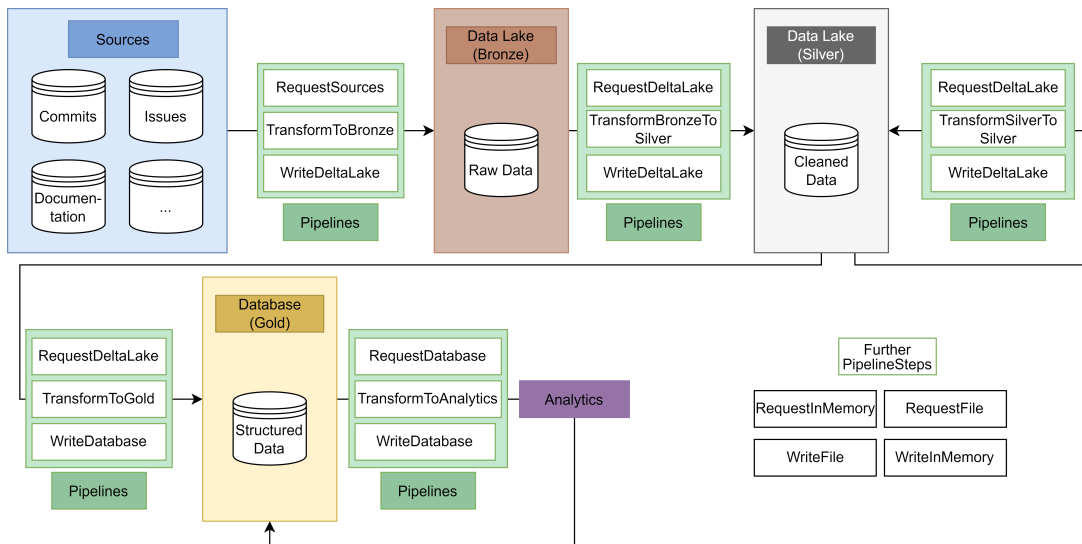


Figure 5.3: Dataflow (‘MECOIS Github Repository’, 2026)

JSON to Bronze

The JSONtoBronzePipeline processes the raw JSON data provided by the sources and writes it unchanged to the Datalake. The data itself isn’t changed in any

way. Each source has an own table.

Bronze to Silver

The `BronzeToSilverPipeline` processes the bronze data in multiple steps to be ready for the analytics.

Flatten table The bronze state cannot represent nested data. (e.g. a Github commit contains an autor and the author itself has multiple properties). In the first step those nested data in columns is split in multiple columns, so that each on only contains a single datapoint.

Drop irrelevant columns Not all data is useful. Some columns (e.g. the URL to the avatar of a Github user) are dropped.

Fix dates Dates are still stored as String. This steps casts these Strings into

Anonymize Data The data is sent to the anonymization service for processing.

add UUID Each table has a way to create an UUID by concatenating some columns and hashing the result.

Silver to Silver

Identities may update in the identity mangement system. The `UpdateIdentityPipeline` updates any identities in the datalake that have been updated in the Sortinghat.

Silver to Gold

There is no single `Silver2GoldPipeline`. The analytics represent the gold to silver pipelines. Each analytic uses the silver data to provide some metric.

Gold to Gold

Some analytics may need some mechanism to update the generated metric. This process is defined as gold to gold transformation.

5.2 Anonymization System

The anonymization system provides the anonymization service, that anonymizes the data according to the GDPR. It provides a client library for sending data and requesting identities.

5.3 Identity Management System

The identity management system is responsible for storing all information about individuals. The project is using the tool Sortinghat from the toolset GrimoireLab by CHAOSS. This application is provided under the GPL-3.0 license.

One individual may have multiple identities on different platforms. The system can pool these different accounts into one individual. These individuals can also be grouped into teams. Multiple teams can form an organization. (‘SortingHat’, 2026) (‘SortingHat Github Repository’, 2026)

5.3.1 Identities

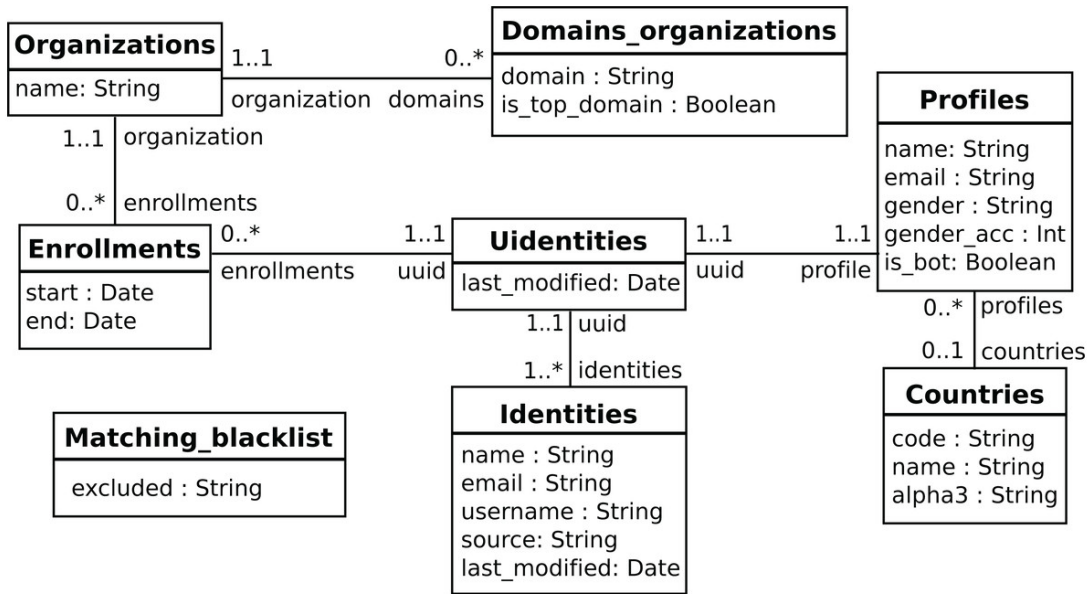


Figure 5.4: SortingHat UML diagram (Dueñas et al., 2021, p. 15)

Figure 5.4 describes how identities are managed in the SortingHat. One individual may have multiple identities on different platforms. The system can pool these different accounts into one individual. One individual has a UUID and is linked to a profile containing personal information. These individuals can also be grouped into organizations. (‘SortingHat’, 2026) (‘SortingHat Github Repository’, 2026)

5.3.2 Frontend Dashboard

The Sortinghat provides a dashboard, where individuals can be merged and associated into teams and organizations.

5. Architecture

Figure 5.5 shows a screenshot of the frontend. Since the screenshot contains real data, the individuals are blacked out. The individuals section can be seen in figure 5.6.

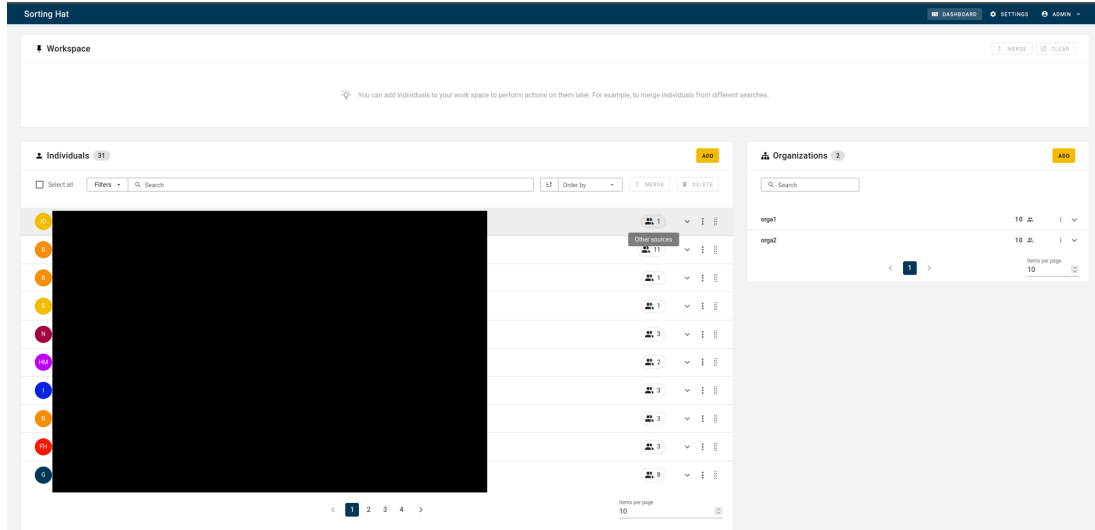


Figure 5.5: SortingHat frontend (individuals are censored for privacy reasons)

5.3.3 API

The Sortinghat also provided an API for automating interactions, like listing, updating and deleting individuals.

Individuals

9

ADD

Select all

Filters

Search

≡

Name

MERGE

DELETE

EM

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Figure 5.6: Sortinhhat individuals section (uncensored) (‘SortingHat’s new interface’, 2026)

6 Design and Implementation

Chapter 5 described the architecture of the MECOIS project and the integration of the anonymization service. This chapter will explain how the service is implemented.

6.1 Compliance with Data Protection Laws

Before implementing the service, the approach to the anonymization has to be discussed.

6.1.1 Reason for an Anonymization Service

According to the BDSG, personal data may only be collected, processed, or used for a specific purpose. It is forbidden to use this data for other purposes. There are only a few specific Exceptions for this rule. (Däubler, 2017, p. 72)

purpose limitation

Since software development tools like Github, Gitlab or Jira contains personal data (e.g. git commit author name or Jira ticket reporter), its use falls under the GDPR. In most cases there is no written agreement on how the data can be used, so it can be assumed that the purpose for this data is to help developers organize their work. Generating productivity measurements for individual employees would be considered a misuse of the data. Däubler (2017, pp. 266–267) explicitly states that data that consists of operational workflows gathered by electronic systems cannot be used for measuring productivity of individual employees. (Däubler, 2017, pp. 266–267)

One exception is scientific research. If individuals are part of a scientific research project, their data can be used for this purpose. The data should get pseudonymized or anonymized as fast as possible, in order to minimize the infringement of privacy of the individual members.(Däubler, 2017, pp. 273–274)

Anonymized Data

The BDSG only applies to personal Data, which is defined in Section 3 (1): Personal data consists of specific information about the personal or factual circumstances of an identified or identifiable natural person (data subject). (Däubler, 2017, p. 69)

The DSGVO has a similar approach. In Article 4(1) it defines personal data as any information relating to an identified or identifiable natural person (hereinafter referred to as the “data subject”). A natural person is considered identifiable if they can be identified, directly or indirectly, in particular by association with an identifier such as a name, an identification number, location data, an online identifier, or one or more factors specific to the physical, physiological, genetic, mental, economic, cultural, or social identity of that natural person. (Däubler, 2017, p. 69)

Unlike anonymized data, where re-identification is impossible, pseudonymized data is still traceable to the original person. therefore pseudonymized data has to comply with the DSGVO and BDSG. Anonymized data is not mentioned in the DSGVO because, since it does not relate to specific individuals, it is not subject to the DSGVO. (Däubler, 2017, p. 69)

Consequences for MECOIS

For the MECOIS project the exception for scientific research is very important regarding the development of the project. As long as MECOIS is a scientific research project, the data can be used to generate productivity measurements as long as the process serves scientific research purposes. But the data still should be pseudonymized or anonymized.

Regarding the final product, MECOIS must anonymize the Data. Otherwise it is illegal to produce productivity metrics.

6.1.2 Approaches for Pseudonymization

6.1.3 Approaches for Anonymization

6.1.4 Deciding on an Approach

6.2 Anonymization Server

6.2.1 Implementing the Server

6.2.2 Implementing the Pseudonymization

TODO: überschrift mit der eigentlichen Methode austauschen

6.2.3 Implementing the Anonymization

TODO: überschrift mit der eigentlichen Methode austauschen

6.2.4 Deploying the Server

Docker Container

7 Evaluation

lorem ipsum

8 Conclusions

lorem ipsum

8. Conclusions

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